Is It the Model or the Reading?
A Fast-Weight View of ViT Attention Maps

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Abstract

Deep-layer attention maps of vision transformers (ViTs) often look poor: scattered, with much of the score on a few background patches. The usual remedy changes the model. We ask how much of the problem lies in how the map is read. A long-standing view from the fast-weight literature treats an attention head as a two-layer network whose hidden units are the tokens. With it, we state three premises behind the common received-attention map, measure each per layer on DeiT-S, DeiT-B and DINO ViT-S, and test a correction for each against a matched control. Two premises fail in deep layers. The diagnosis predicts the outcome of the correction for two of the three premises: contribution weighting, in practice the value-norm weighting of Kobayashi et al. and Zhu et al., improves the maps exactly in the models where its premise fails (object enrichment $+0.038$ and $+0.065$). For the third it fails: relocating scores hurts everywhere. On DeiT-S, two model-side fixes do no better. Part of what makes deep maps look poor is how we read them.

1 Introduction

Attention maps are widely used to show what a vision transformer (ViT) [12] looks at [1, 8, 9]. In deep layers they often look poor: the score is scattered, much of it on a few background patches [10]. The usual remedy changes the model: register tokens absorb the stray attention [10, 17]. We ask instead: *how much of the problem lies in how the map is read?*

A map is a model plus a reading rule, and the rule is rarely examined. We view an attention head as a two-layer network that is written anew for each image and has the tokens as its hidden units (Fig. 1), a long-standing view from the fast-weight literature (Section 2). The common *received-attention* map then reads the hidden units of this network, and it rests on three premises that can be tested: activations are specific to the image, unit j still stands for patch j, and a large activation is a large contribution (Section 3). On DeiT-S, DeiT-B [24] and DINO ViT-S [8], we check that the model uses the units that attention marks, measure each premise per layer, test a correction for it against a matched control, and ask whether the diagnosis predicts where the correction helps (Section 4). Our contributions are:

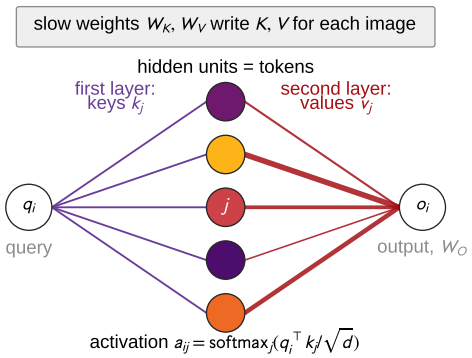


Figure 1: The lens. For one image, an attention head is a two-layer network: the slow weights W_K, W_V write its weights (keys, values), and each token is a hidden unit.

(1) **Premises.** We use the fast-weight view to state three testable premises of the attention-map reading; to our knowledge, it has not been used to test how attention maps are read.

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(2) **Protocol.** For each premise: a diagnostic, a correction, a matched control and centre-blind metrics, so that the diagnosis predicts where the correction should and should not help.
(3) **Findings.** Two premises fail in deep layers. The prediction holds for two: value-norm weighting [18, 29] helps where its premise fails. It fails for the third: relocation hurts everywhere.

2 Related work

How attention maps are read and scored. Whether attention explains predictions is debated [16, 26]. Later readings add layers (rollout [1]), gradients and the class (GAE [9]), vector norms [18, 29] or statistical filtering [3]. Scoring has traps: plausible maps need not depend on the model [2], deletion scores depend on the occlusion strategy [5], and a centre bias makes random saliency look trained [19]. These works compare readings by their *results*; none tests their premises.

Why deep ViT maps degrade. Deep maps put much attention on high-norm tokens at uninformative background patches, which registers absorb [10, 17]; such tokens resemble heads that learn to do nothing [6]. Tokens also mix with depth; in language models their identity is largely kept but fades gradually [7]. These effects are repaired in the model, not separated from flaws of the reading.

Attention as a fast-weight network. Fast weights are weights that one network writes for another [23]; linear attention is such a fast-weight program [22], and softmax attention is the update rule of a modern Hopfield memory [21]. This view of a head as a network is not ours and has mostly guided model design; MiTA [25], with the same view but another goal, makes attention efficient. To our knowledge, it has not been used to test how attention maps are read. We use this third thread to state the premises behind the first and to ask whether the second's problems lie in model or reading.

3 Method

The lens. In layer ℓ, the normalised tokens $x_1, \dots, x_{N+1}$ (CLS and $N{=}196$ patches) enter H heads. Head h forms $q_i^h{=}W_Q^h x_i$, $k_j^h{=}W_K^h x_j$ and $v_j^h{=}W_V^h x_j$, and the block adds to token i

$$o_i = \sum_h W_O^h \sum_j a_{ij}^h \, v_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j\big(q_i^{h\top} k_j^h / \sqrt{d}\big). \tag{1}$$

For a fixed image this is a two-layer network applied to q_i: token j is a hidden unit with input weights k_j^h, activation a_{ij}^h and output weights $W_O^h v_j^h$. As the softmax makes units compete, the network is a guiding approximation, not an exact equivalence. *Received attention* sums each unit's activation over queries and heads, $s_j = \sum_h \sum_i a_{ij}^h$, and draws it at patch j; token pruning ranks tokens by the same score [15]. As our question concerns deep layers, maps sum s_j over layers 9–12 (the last third), a range set before the experiments (per-layer values: Fig. 2, App. B).

From reading units to premises. If a head is a network, an attention map *reads its hidden units*, and three lessons about reading units each give a premise of the received-attention map, with a diagnostic, a correction and a matched control: an equally large change of the map without the premise's information. A correction counts only if it beats its control, and it should help where the premise fails and change little where it holds. *Specificity.* What is there for every input, such as a centre bias [19], says nothing about this input. Premise: activations are specific to the image. Diagnostic: the share of a map's spatial variance explained by the mean map $\bar{s}$ of held-out images. Correction: $s - \bar{s}$. Control: subtracting a centred Gaussian of the same spread. *Location.* What a unit stands for must be measured, not read off its index [4]. Premise: unit j still stands for patch j. Diagnostic: the accuracy of a linear probe that predicts the patch index of token x_j. Correction: probe relocation, $r_p = \sum_j s_j P(p \mid x_j)$, moves each score to where the probe places its unit. Control: a Gaussian blur of equal mean displacement. *Contribution.* A large activation is not a large effect: output weights count too, as in class activation maps [28]. Premise: a large activation is a large contribution. Diagnostic: the share of s on high-norm tokens (block-input norm above three times the image median). Correction: $c_j = \sum_h \sum_i a_{ij}^h \|W_O^h v_j^h\|_2$, activation times output-weight norm; the norm-based weighting of Kobayashi et al. [18] applied to received attention. Control: zeroing them.

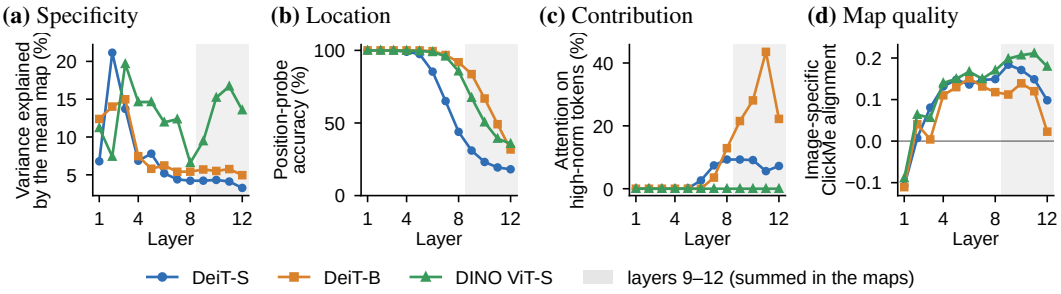


Figure 2: Diagnosis per layer: (a)–(c) the three premises; (d) single-layer map quality. Bands: 95% intervals.

Metrics. A centre prior can inflate scores: it makes random saliency maps look trained [19], and in our data a centred Gaussian that ignores the image has a Spearman correlation of 0.41 with human ClickMe maps. Our two main metrics are centre-blind. *Same-ring object enrichment* compares the object rate (ImageNet-S masks [14]) of the top-scoring patches with that of all patches equally far from the centre, so radial maps score zero. *Image-specific ClickMe alignment* [13, 20] is the Spearman correlation with the image's own human map minus that with another image's, so image-independent maps score zero. The third metric, *deletion faithfulness* [5], is not centre-blind (a centred Gaussian scores 0.099–0.140). A paired difference is *higher* or *lower* only if its 95% bootstrap interval excludes zero and it exceeds a margin (0.02; 0.05 for deletion); else it is *comparable*.

4 Experiments

Setup. We use frozen DeiT-S and DeiT-B [24] (supervised) and DINO ViT-S/16 [8]; the model-side fixes are DeiT-S with a test-time register [17] and a DeiT-S trained with top-k attention (MiTA [25]). We evaluate on 5,000 ImageNet validation images [11] (execution test), 12,419 ImageNet-S images [14] (2,000 for deletion) and 17,408 ClickMe maps [13, 20]; 4,000 held-out images give probes, mean maps and blur widths (App. A). The metrics, margins, comparisons with controls and the predictions of Table 2 were fixed before the experiments; later analyses are marked *post hoc*.

Attention marks the units the model uses. In every head and layer, each query reads only the k units that its region's mean query scores highest (16 regions). With k=16 of 196 units, the models keep 72%, 83% and 61% of their top-1 predictions (DeiT-S, DeiT-B, DINO ViT-S), against 9%, 30% and 4% for random units of the same number; the gap holds for every k from 4 to 128 (App. B).

Two premises fail in deep layers. *Specificity largely holds* (Fig. 2a): in layers 6–12 the mean map explains 3–6% of a map's spatial variance in the supervised models and at most 17% in DINO ViT-S, and in layers 9–12 it is 82–98% radial: a centre bias. *Location fails in all three models* (Fig. 2b): probe accuracy falls from close to 100% in layers 1–5 to 18%, 32% and 36% at layer 12 (chance: 0.5%). *Contribution fails only in the supervised models* (Fig. 2c): high-norm tokens get up to 44% of the attention in DeiT-B and about 9% in DeiT-S but none in DINO ViT-S, as in [10], and their value vectors are only 0.29 and 0.23 times as long as those of other patches.

Only contribution weighting improves the map (Table 1). *Contribution weighting* is higher than received attention on both centre-blind metrics in DeiT-S and DeiT-B and comparable on deletion. At layer 12, where map quality drops (Fig. 2d), it recovers 31–62% of the drop on both metrics. It beats its control on enrichment in both models and on ClickMe in DeiT-S. Weighting by $\|v_j\|$ alone does the same (post hoc; App. B), so in practice it is value-norm weighting [18, 29]. *Mean-map subtraction* gains only on DINO ViT-S ClickMe, where its control gains as much, and is lower in two other cells. *Probe relocation* is lower in all nine cells; in DeiT-S and DINO ViT-S it is lower even than its matched blur on all three metrics (not in DeiT-B), which suggests the harm lies in where the score is sent, not how far. We expected the corrections to add up; together they inherit this harm instead (higher in one cell, lower in seven). Neither model-side change (DeiT-S only) is higher.

Does the diagnosis predict the correction? Each prediction of Table 2 could have failed. For *contribution* it holds: the correction helps in the two models where the premise fails and changes

Table 1: Readings over layers 9–12 (controls indented): same-ring object enrichment (Enr.), image-specific ClickMe alignment (Click.), deletion faithfulness (Del.). $^{\uparrow}/^{\downarrow}$: higher/lower than received attention (model-side rows: vs. DeiT-S). GAE, which also uses gradients and the class, is an untested reference. Intervals: App. B.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	$0.178^{\uparrow}$	$0.190^{\uparrow}$	0.253	$0.146^{\uparrow}$	$0.165^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	$0.121^{\uparrow}$	$0.148^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077	$0.087^{\downarrow}$	0.144	0.201	$0.252^{\uparrow}$	$0.233^{\downarrow}$
centre-Gaussian subtraction	0.133	0.168	$0.167^{\downarrow}$	$0.056^{\downarrow}$	$0.062^{\downarrow}$	$0.024^{\downarrow}$	0.200	$0.253^{\uparrow}$	$0.232^{\downarrow}$
Probe relocation	$0.041^{\downarrow}$	$0.084^{\downarrow}$	$0.076^{\downarrow}$	$0.027^{\downarrow}$	$0.050^{\downarrow}$	$0.051^{\downarrow}$	$0.084^{\downarrow}$	$0.143^{\downarrow}$	$0.141^{\downarrow}$
matched blur	$0.095^{\downarrow}$	0.152	$0.181^{\downarrow}$	$-0.008^{\downarrow}$	$0.068^{\downarrow}$	$0.097^{\downarrow}$	0.189	0.226	0.287
All three corrections	$0.059^{\downarrow}$	$0.123^{\downarrow}$	$0.076^{\downarrow}$	0.079	$0.137^{\uparrow}$	$0.085^{\downarrow}$	$0.088^{\downarrow}$	$0.177^{\downarrow}$	$0.090^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	$0.100^{\downarrow}$	$0.115^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Table 2: Does the diagnosis predict the correction? Centre-blind metrics. *Only as much as its control.

Premise (correction)	Premise fails in	Correction helps in	Predicted
Specificity (mean-map subtraction)	none	none*	✓
Location (probe relocation)	all three models	none (lower in all nine cells)	✗
Contribution (contribution weighting)	DeiT-S, DeiT-B	DeiT-S, DeiT-B (not DINO ViT-S)	✓

nothing in DINO ViT-S, where it holds. Two post-hoc tests agree: within a model, images with more attention on high-norm tokens gain more, and shuffling the weights among each head's tokens removes the gain (App. B). For *specificity*, a small and radial image-independent part predicts no gain beyond its control, as found. For *location* the prediction fails: relocation is harmless in layers 1–4, where the premise holds, but harmful in deep layers. Post hoc, we found no support for four explanations (App. C): far probe errors, a fixed position preference, a weak probe (a stronger MLP probe gains 2–7 points and still harms), and harder-to-place global tokens. Why it harms is open.

5 Discussion

Interpretation. Two lessons carry over. *Activation is not effect:* high-norm tokens draw attention but have short value vectors, like heads that do nothing [6]; weighting by output weights partly repairs the map without changing the model, and neither model-side change improves it. *Index is not identity:* deep units no longer stand for their patch, yet moving scores back does not repair the map.

Limitations. We study three classifiers at S/B scale, each trained once, and only received attention: CLS-attention maps, larger models and models trained with registers are untested, and the model-side fixes were compared on DeiT-S only (App. A). ClickMe images were very likely part of the models' training data. The gains are modest, hard to see in one image (App. D), and below GAE's.

Ethical considerations. We use public data and checkpoints; the images include people and known biases. An attention map, corrected or not, is only partial evidence about a model.

6 Conclusion

An explanation map is a model plus a reading rule. For ViT attention maps, the fast-weight view yields three premises: two fail in deep layers, and correcting the reading partly repairs one, where the diagnosis predicts. We tested only ViT attention maps; one can ask the same of any method that maps internal values to input positions. *Future work:* larger and register-trained models, CLS attention, recovering location. *Broader impact:* premise tests need no retraining and curb over-trust.

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A Experimental details

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs, taken from timm [27]. *DeiT-S and DeiT-B* [24]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224). *DeiT-S with a test-time register* [17]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [25] (25 summary slots, $k = 25$), with its own preprocessing; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side fixes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [11]. *ImageNet-S* [14]: all 12,419 validation images with segmentation masks; enrichment is defined for 12,150 of them (12,144 for DINO ViT-S), and deletion uses a random subset of 2,000. *ClickMe* [13, 20]: 17,408 images with human importance maps, which we smooth with a Gaussian. *Held-out set*: 4,000 ImageNet validation images that are not evaluated anywhere; 2,000 train the position probes, and the other 2,000 give the mean maps, the probe accuracies and the blur widths. While building this set we found that only 2 of 4,002 candidate validation images matched a ClickMe image, so the ClickMe images are very likely ImageNet training images, which all three models have seen.

Corrections and matched controls. All are applied per layer and then summed over layers 9–12; queries run over all 197 tokens and maps cover the 196 patches. *Mean-map subtraction* subtracts the model's mean received-attention map over the 2,000 held-out images. Its control subtracts a centred isotropic Gaussian ($\sigma = 3.5$ patches) scaled to the spatial standard deviation of that mean map; it is exactly radial. *Probe relocation* uses one linear 196-way classifier per model and layer (36 in all) on the standardised tokens after the block's first layer norm (Adam, learning rate 0.01, 30 epochs) and computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, which keeps the total score. Its control is a Gaussian blur whose width is set so that it moves the score as far, in mean squared displacement, as relocation does on held-out tokens of that layer (0.7–2.9 patches). *Contribution weighting* computes $c_j = \sum_h \sum_i a_{ij}^h \|W_O^h v_j^h\|_2$, where W_O^h is the head's slice of the output projection; $a_{ij}^h \|W_O^h v_j^h\|$ is the per-head norm of Kobayashi et al. [18]. A *high-norm token* is a patch whose block input has a norm above three times the image's median patch norm in that layer; the control sets received attention to zero at these patches. *All three corrections*: contribution weighting, then probe relocation, then subtraction of the held-out mean of the resulting map.

Metrics. *Same-ring object enrichment.* A patch is an object patch if at least half of its pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. A purely radial map scores zero in expectation. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the human map of another image. A map that does not depend on the image scores exactly zero. *Deletion faithfulness.* Patches are removed in the order of the map by dropping their tokens; at budgets of 0 to 147 patches we record whether the top-1 class is unchanged. The score is the symmetric relevance gain of Blücher et al. [5]: the area under the least-important-first curve minus that under the most-important-first curve. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps score exactly zero on the latter); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired bootstrap over images (1,000 resamples, 95% percentile intervals). A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its size exceeds 0.02 (0.05 for deletion), and *comparable* (*c*)

otherwise. Definitions, margins and the comparisons of Table 1 were fixed before the experiments; later analyses are marked post hoc.

Compute and code. All experiments took about 19 GPU hours on NVIDIA L4 GPUs. Code and result tables: https://github.com/hhuang999/vit-fastweight-lens.

B Additional results

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer and head, each region's mean query is scored against all keys, and its k highest-scoring keys form the region's reading set (the CLS query keeps its own top-k set). The sets come from one pass of the unmodified model. We then rerun the model with every query restricted to its region's set, in all 12 layers at once, and record whether the top-1 class is unchanged. Figure B.1 shows that top-k sets beat random sets of the same size at every k from 4 to 128 in all three models, and that agreement rises with k.

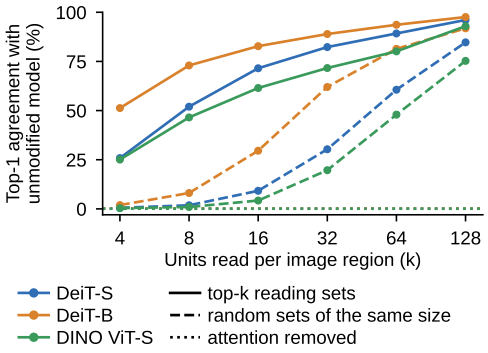


Figure B.1: Top-1 agreement with the unmodified model on 5,000 ImageNet validation images when queries read only top-k units (solid), random units (dashed), or nothing (dotted; all models evaluated with a 0.875 crop). 95% intervals are at most ±1.5 points.

Diagnostics per layer. Table B.1 lists the values plotted in Fig. 2. Two numbers of the main text are not in the table. First, in layers 9–12, 82–98% of the variance of the mean map is explained by the distance from the grid centre. Second, averaged over layers 9–12, the mean value-vector norm $\|v_j\|$ of high-norm tokens is 0.29 (DeiT-S) and 0.23 (DeiT-B) times the median of the other patches (measured on 11,919 ImageNet-S images).

Table B.1: Diagnostics per layer (S = DeiT-S, B = DeiT-B, D = DINO ViT-S): share of the spatial variance of single-layer received attention explained by the held-out mean map; accuracy of the linear position probe (chance 0.5%); share of received attention on high-norm tokens; image-specific ClickMe alignment of the single-layer map.

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Differences behind Table 1. Table B.2 gives the paired differences with their intervals. The values in Table 1 have 95% intervals of at most ±0.005 on enrichment and ClickMe alignment and ±0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ $[0.011, 0.013]$ on enrichment, $+0.013$ $[0.012, 0.013]$ on ClickMe and 0.000 $[-0.005, 0.004]$ on deletion for the

test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training.

Table B.2: Paired differences, layers 9–12: mean with 95% paired-bootstrap interval and verdict (*h* higher, *c* comparable, *l* lower). Top: against received attention. Bottom: each correction against its matched control.

Model	Same-ring enrichment	ClickMe alignment	Deletion faithfulness
Contribution weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Contribution weighting − zero high-norm tokens (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (Fig. 2d). Applied to layer 12, contribution weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Tests of the contribution account. Contribution weighting should work by down-weighting tokens that receive attention but contribute little. One consequence was predicted in advance, and two further tests were added post hoc. *No gain without high-norm tokens (predicted).* In DINO ViT-S, where no attention falls on high-norm tokens, every difference from received attention is at most 0.004 in size (Table B.2). *More high-norm attention, more gain (post hoc).* Across the images of a model, the gain grows with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *The weights must match the tokens (post hoc).* Against the same weights shuffled among the patches of each head, contribution weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. Finally, also post hoc, weighting by $\|v_j^h\|$ instead of $\|W_O^h v_j^h\|$ is comparable in every cell (differences -0.010 to 0.000): in practice the correction is the value-norm weighting of Kobayashi et al. [18] and Zhu et al. [29].

C Why does probe relocation fail?

Relocation lowers every deep-layer cell (Table 1). Two planned comparisons narrow the problem. Where the premise holds, relocation is harmless: at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. And in DeiT-S and DINO ViT-S it is lower than a blur

that moves the score equally far (Table B.2). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none is supported. In particular, the probe is not too weak, and its errors lie mostly next to the true patch; why it still harms the map is open.

Table C.1: Four explanations for the failure of probe relocation, and their tests (post hoc). Cells are the two centre-blind metrics in the three models.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant, wrong patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies within the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Accuracy at layer 12 rises by only 1.8, 3.8 and 6.7 points; relocation with the MLP probe is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

D Example maps

Figure D.1 shows DeiT-S maps for two ClickMe images that were drawn at random, not selected. Received attention and contribution weighting look much alike, while probe relocation spreads the score over many more patches. The gains in Table 1 are averages of small changes in how each map ranks its patches, which single maps do not show.

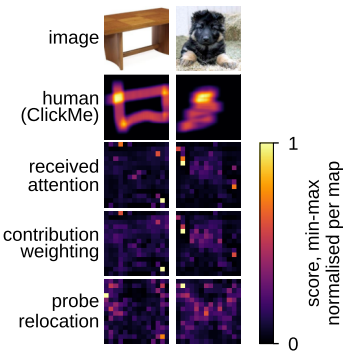


Figure D.1: DeiT-S on two randomly drawn ClickMe images, layers 9–12; each map is min–max normalised.

E Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,N,N] attention a_ij^h (queries i, keys j); v: [B,H,N,d] values; w_o: [D,H,d] W_O by head
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)                  # W_O^h^T W_O^h
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v).clamp_min(0).sqrt()
    received = p.sum(2)                                            # sum over queries i
    s = received.sum(1)[:, prefix:]                                # received attention s_j
    c = (received * length).sum(1)[:, prefix:]                     # contribution weighting c_j
    return s, c

def relocate(s, probs):                  # probe relocation r_p = sum_j s_j P(p | x_j)
    return torch.einsum('bj,bjp->bp', s, probs)
```